Advances and New Challenges on Branch-and-Prune Algorithm

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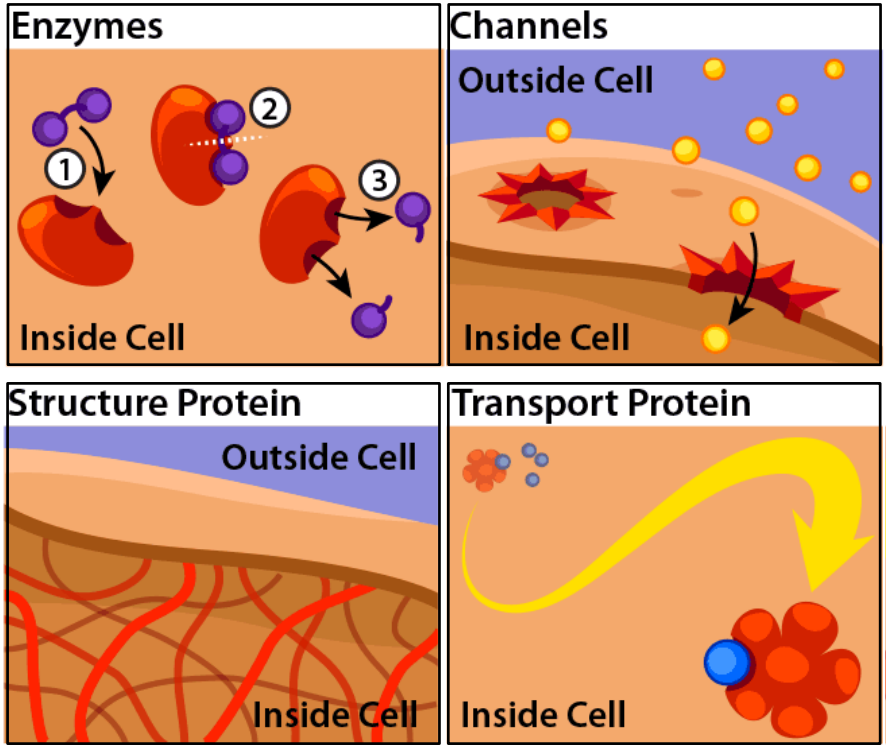
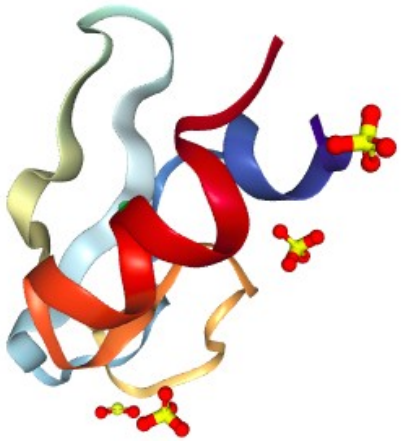
DIMACS'19

Workshop on Optimization
in Distance Geometry

"***Proteins are the machines and building blocks of living cells***. If we compare a living body to our world, each cell corresponds to a town, and the proteins are the houses, bridges, cars, cranes, roads, airplanes, etc. ***There are huge numbers of different proteins***, ***each one performing its specific task***." [Neumaier97]

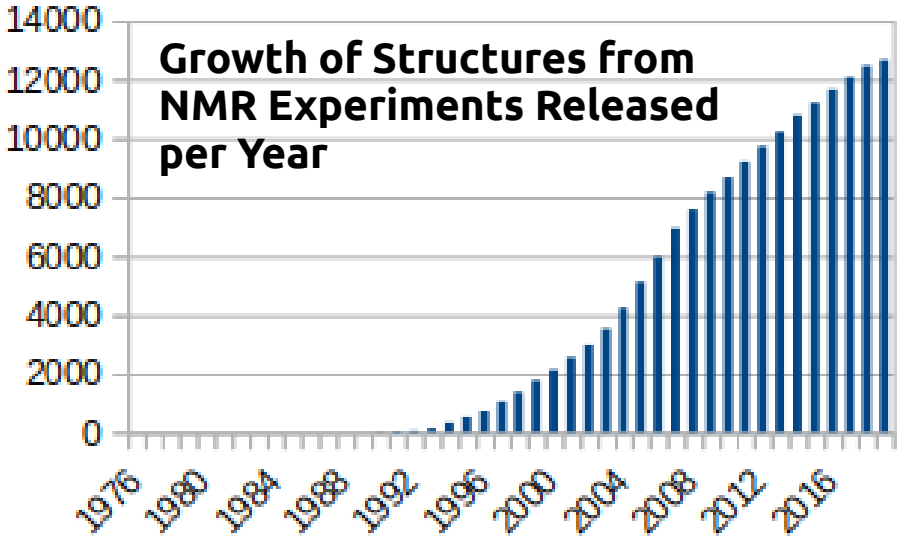




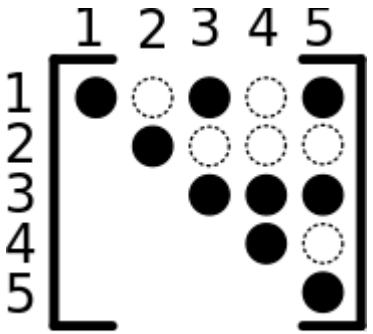
Geometry ⟺ **Function**

PDB Data Distribution by Experimental Method

X-Ray	128,043
NMR	11,109
Electron Microscopy	2,416
Other	257
Multi Method	137
Total	**141,962**



NMR does not give atoms location, but info about pairwise distances.



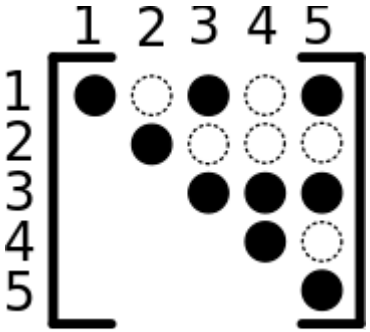
Incomplete (imprecise) EDM

Distance Geometry Problem (DGP_k)

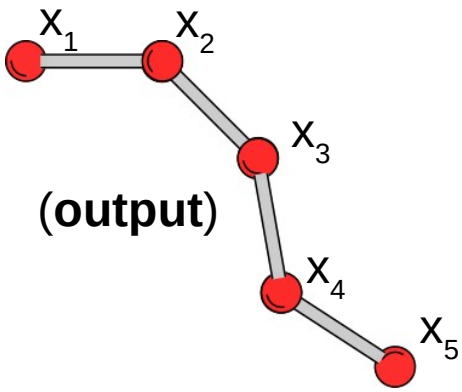
Given a simple weighted undirected graph $G = (V, E, d)$, where $d : E \to \mathbb{R}_+$, find an embedding $x : V \to \mathbb{R}^k$ such that

$$\forall \{i, j\} \in E, \ \| x(i) - x(j) \| = d(i, j)$$

If the vertices are ordered, we can represent a DGP_k instance using an Euclidean Distance Matrix (EDM)



Incomplete EDM (**input**)



(**output**)

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- DGP_k is a NP-Hard problem
  > Reduction to Subset-Sum [Saxe79]

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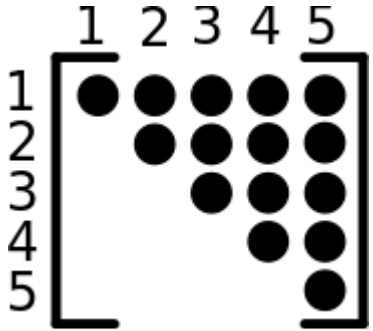

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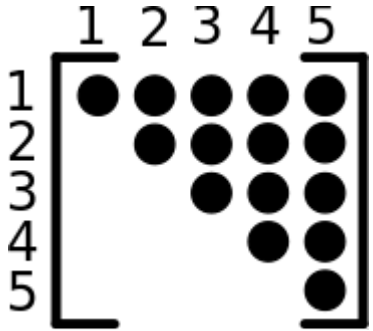

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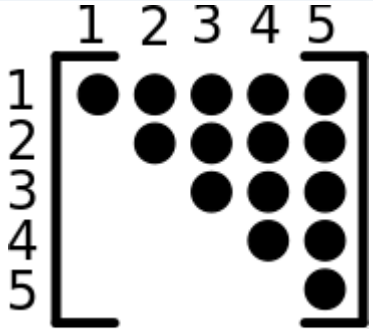
The power of Divide-and-conquer

Instead of solving the entire problem at one step taking $O((kn)^3)$ operations., it's better to ***solve n small problems*** with $O(k^3)$ operations.



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Discretizable Molecular DGP_k ($DMDGP_k$)

We say that a DGP_k instance is ***discretizable molecular*** DGP if there is an order $v_1, v_2, \ldots, v_n$ of V satisfying the following requirements:

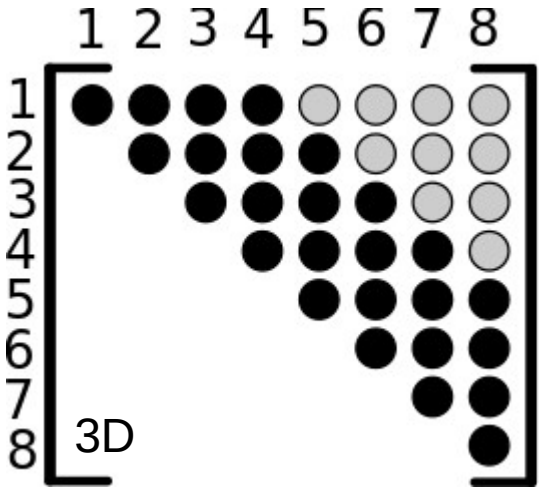
1. *E* contains all cliques on k + 1 consecutive vertices;

2. For all $w \in \{ k+1, \ldots, n \}$,

$$\mathrm{Vol}(\{ d(i,j) : w - k \leq i, j < w \}) > 0$$

(2D: x_{w-1} and x_{w-2} are not overlapped)

(3D: x_{w-1}, x_{w-2}, x_{w-3} are not colinear)



The other distances (gray) may be available

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[Lavor12]

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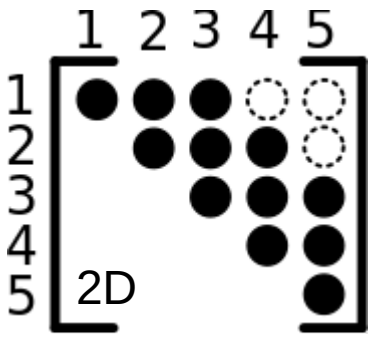
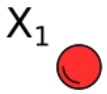
- **The search space of DMDGP$_k$ is finite (discrete), but it can be large (up to $2^{|V|-k}$)**





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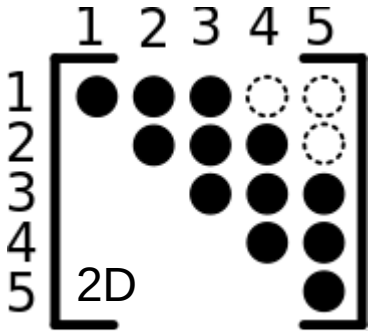

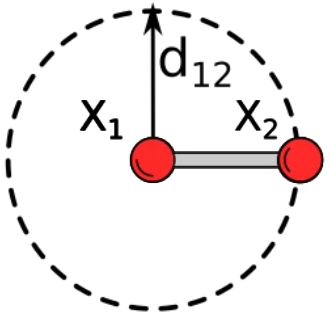

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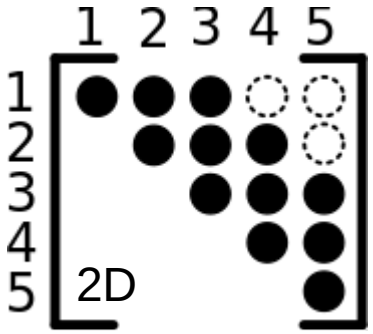

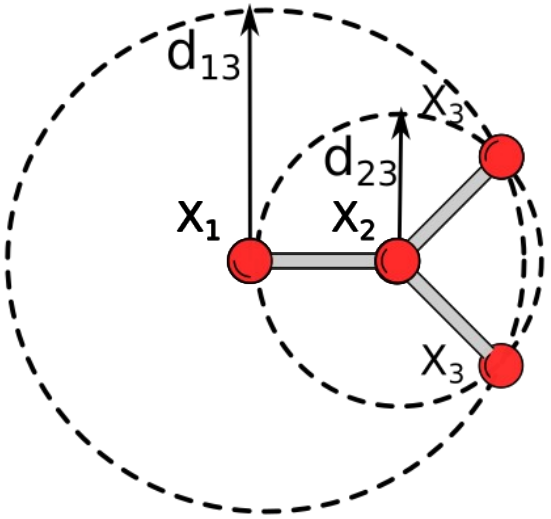

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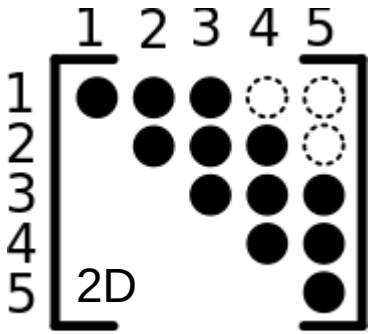

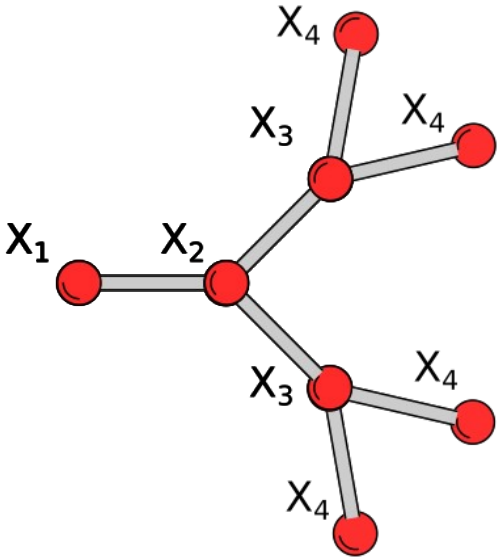

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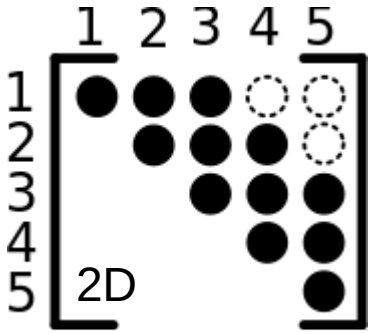

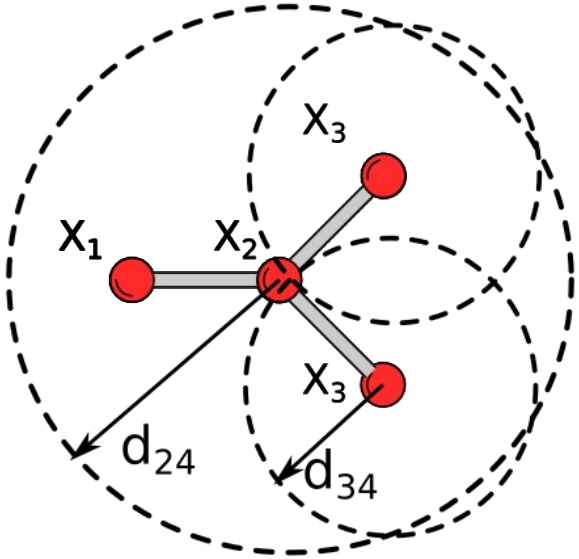

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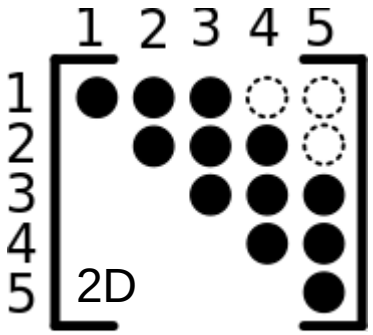

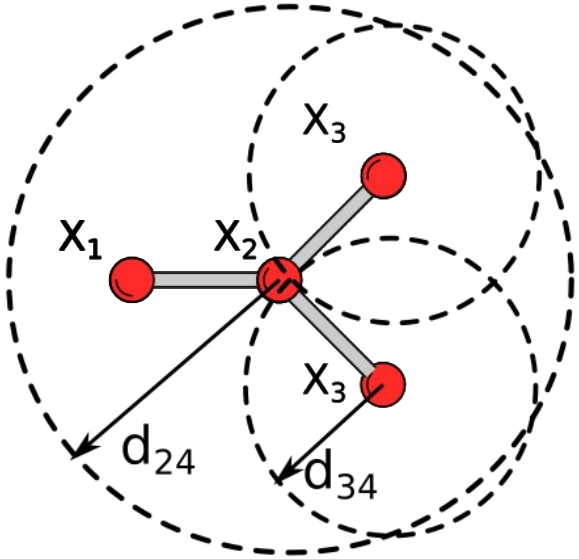

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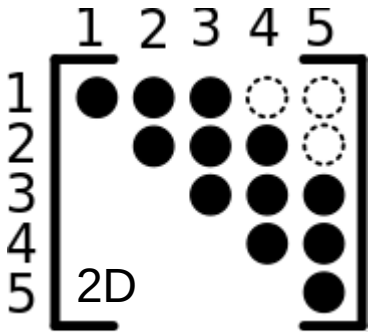

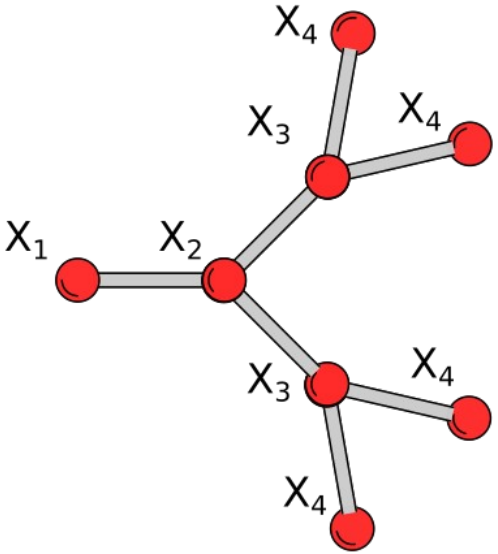

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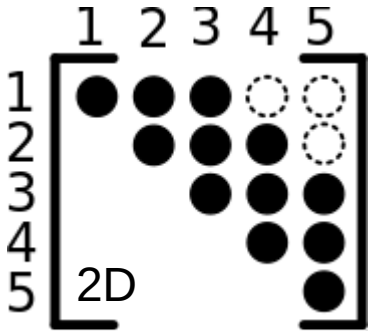
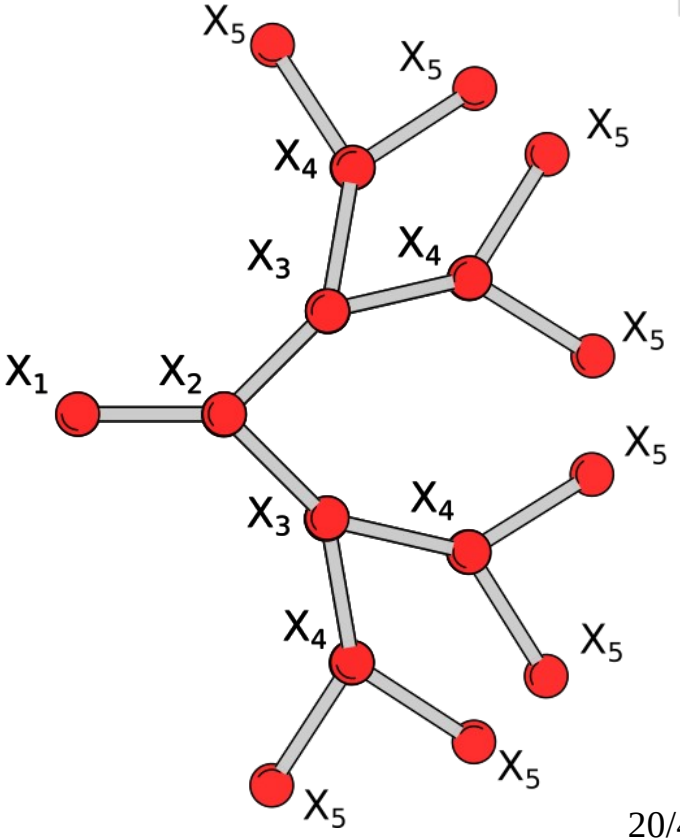
Globally not rigid





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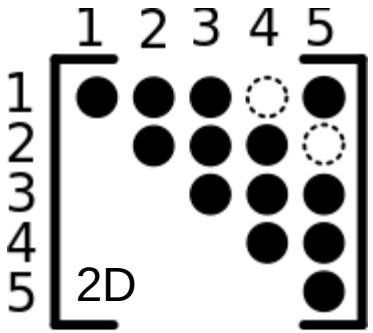

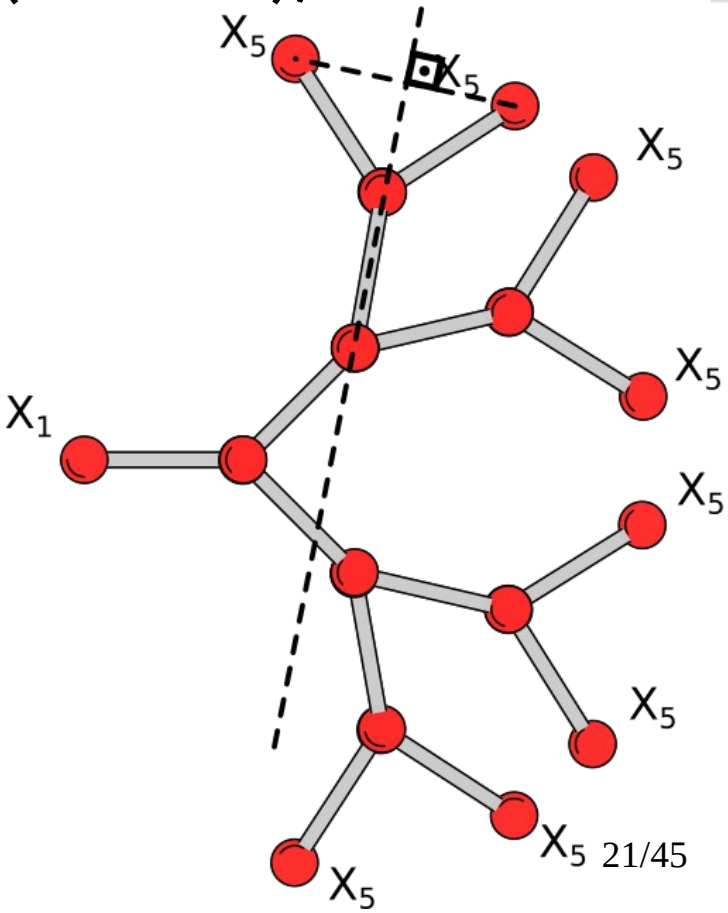

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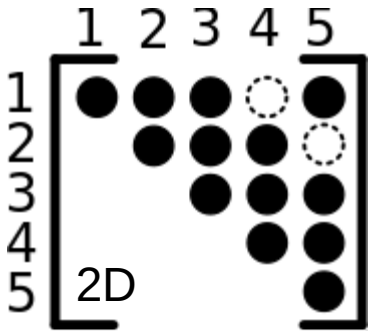

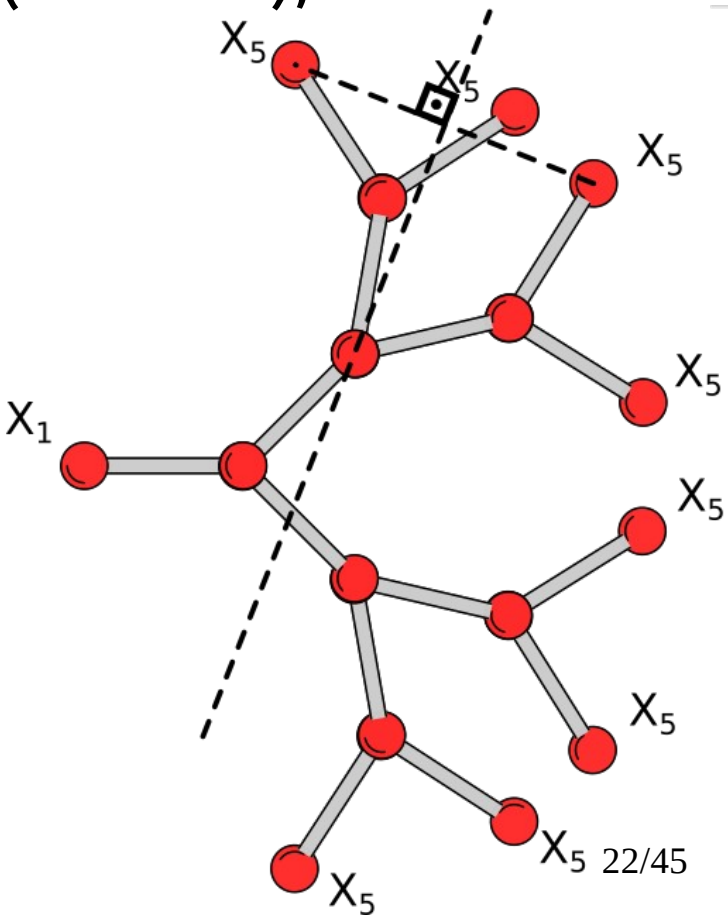

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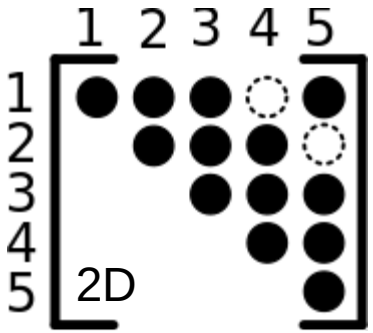

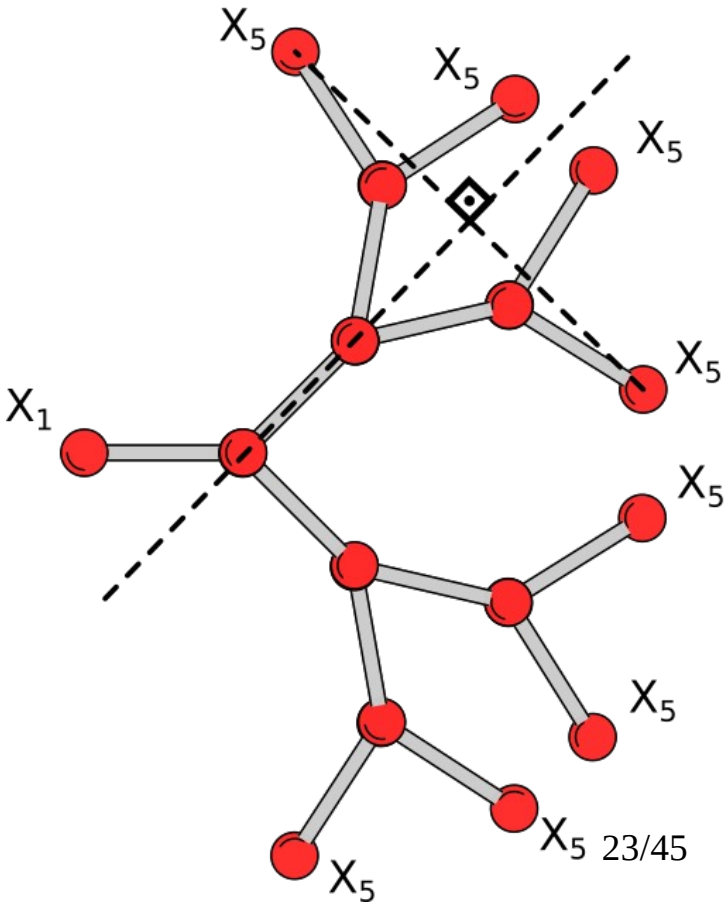

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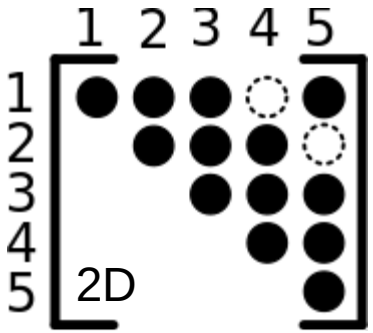

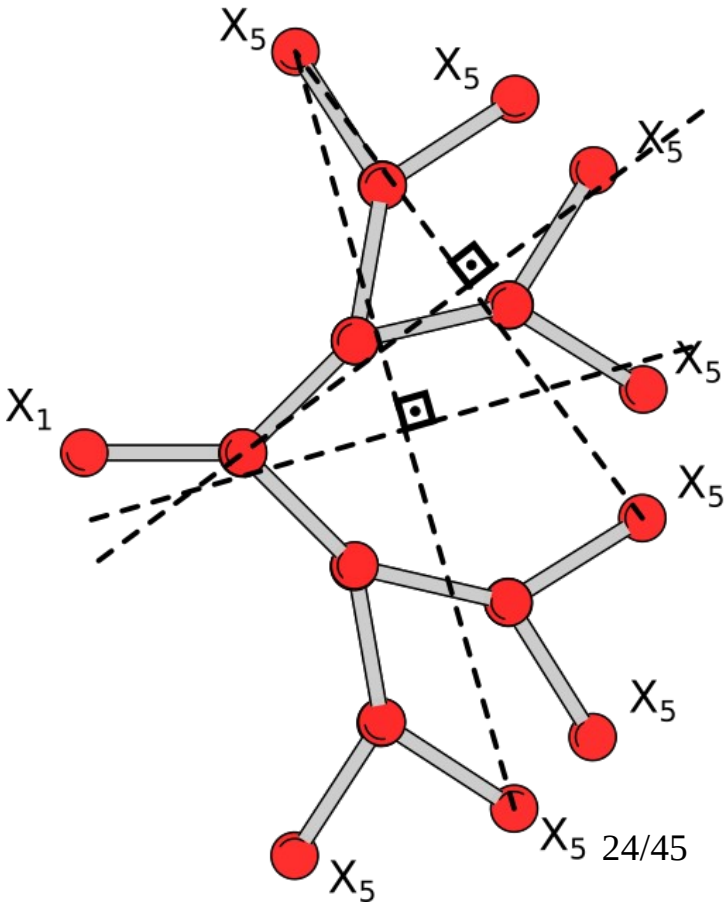

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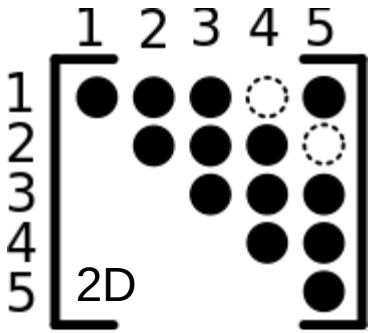

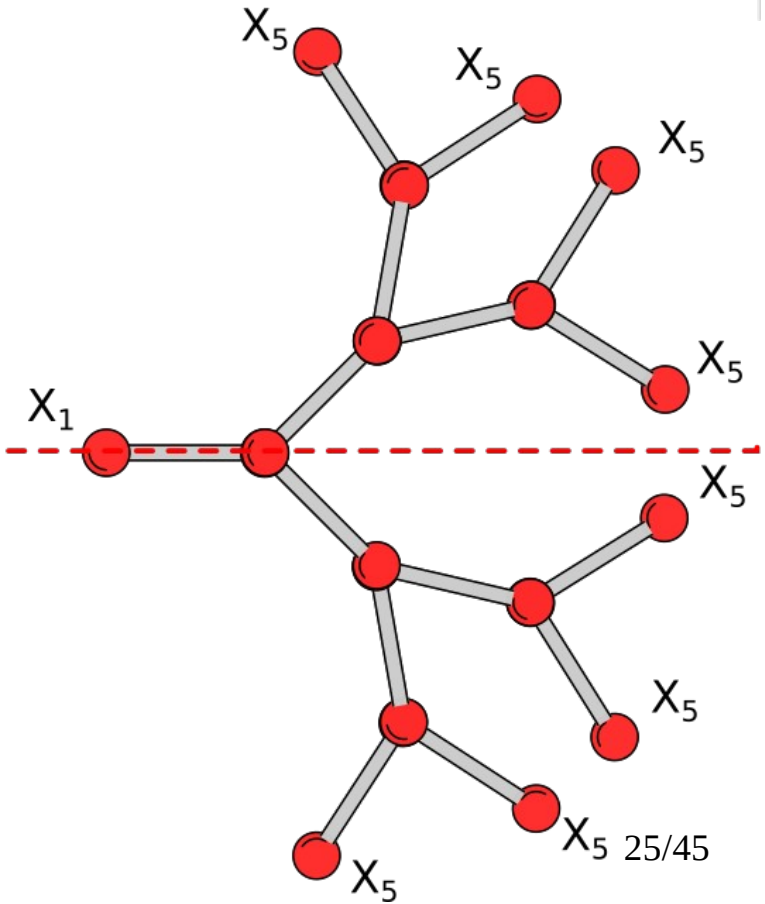

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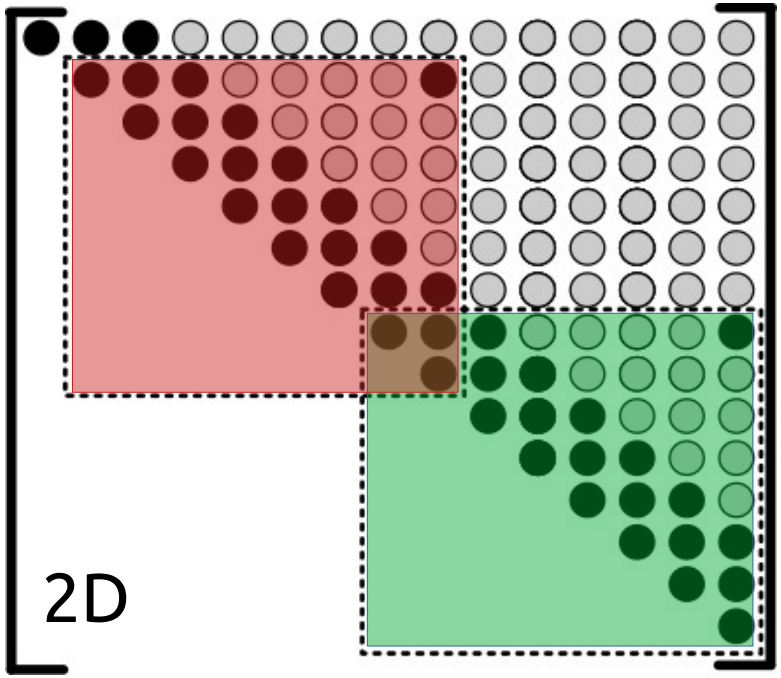
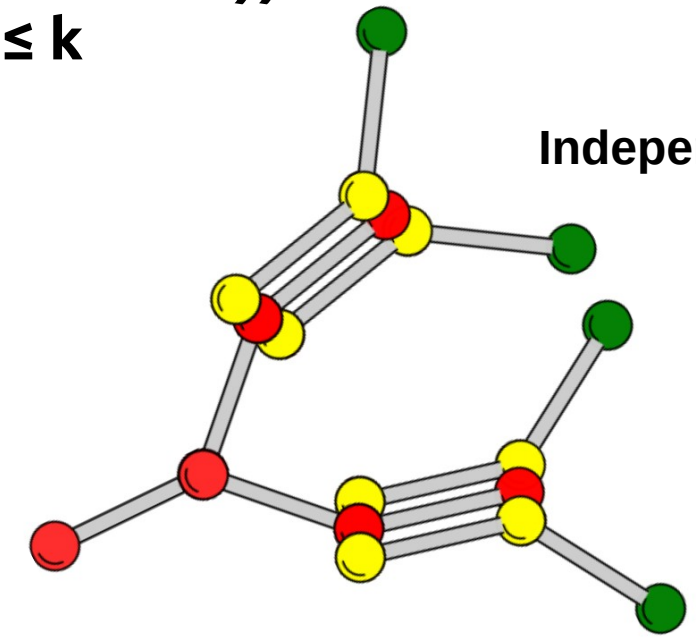
Independent Components (IC) – Symmetries



The ***orientation of the structures*** associated to the distances d(i,j) and d(u, v), where i < j, u < v ***are independent iff***

j – u ≤ k



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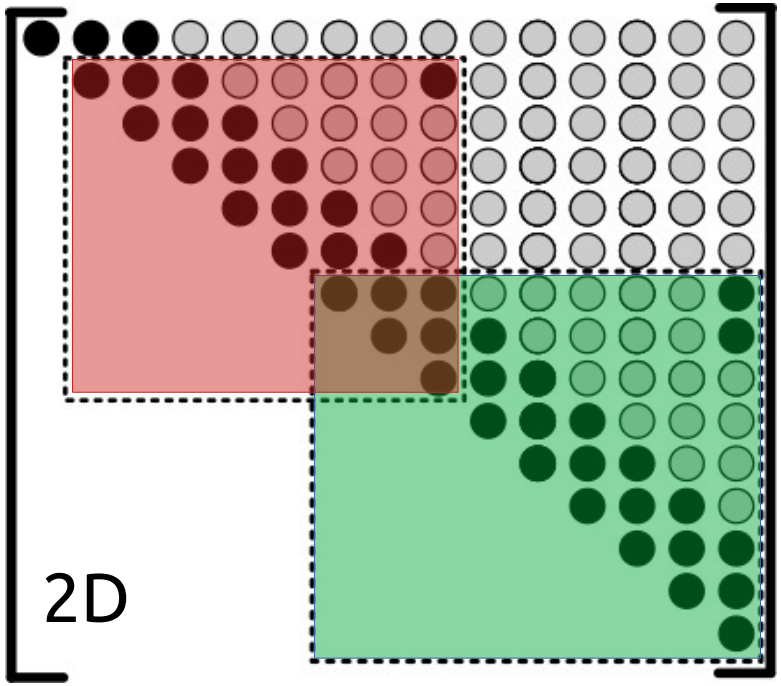
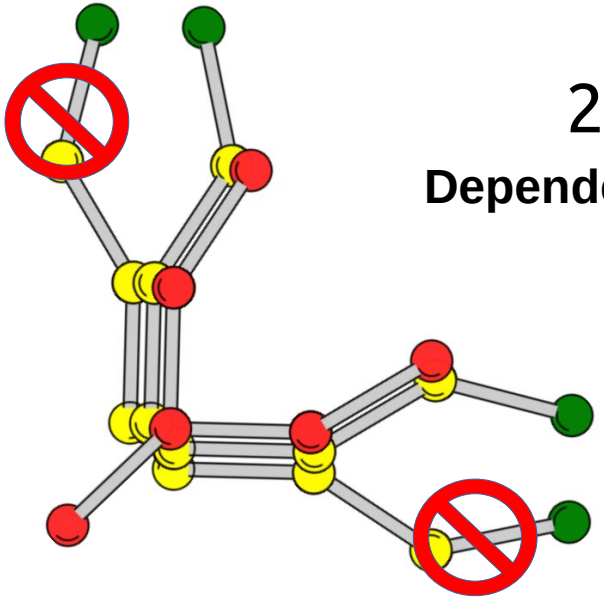


The number of solutions of $DMDGP_k$ is 2^λ, where λ is the number of IC's.



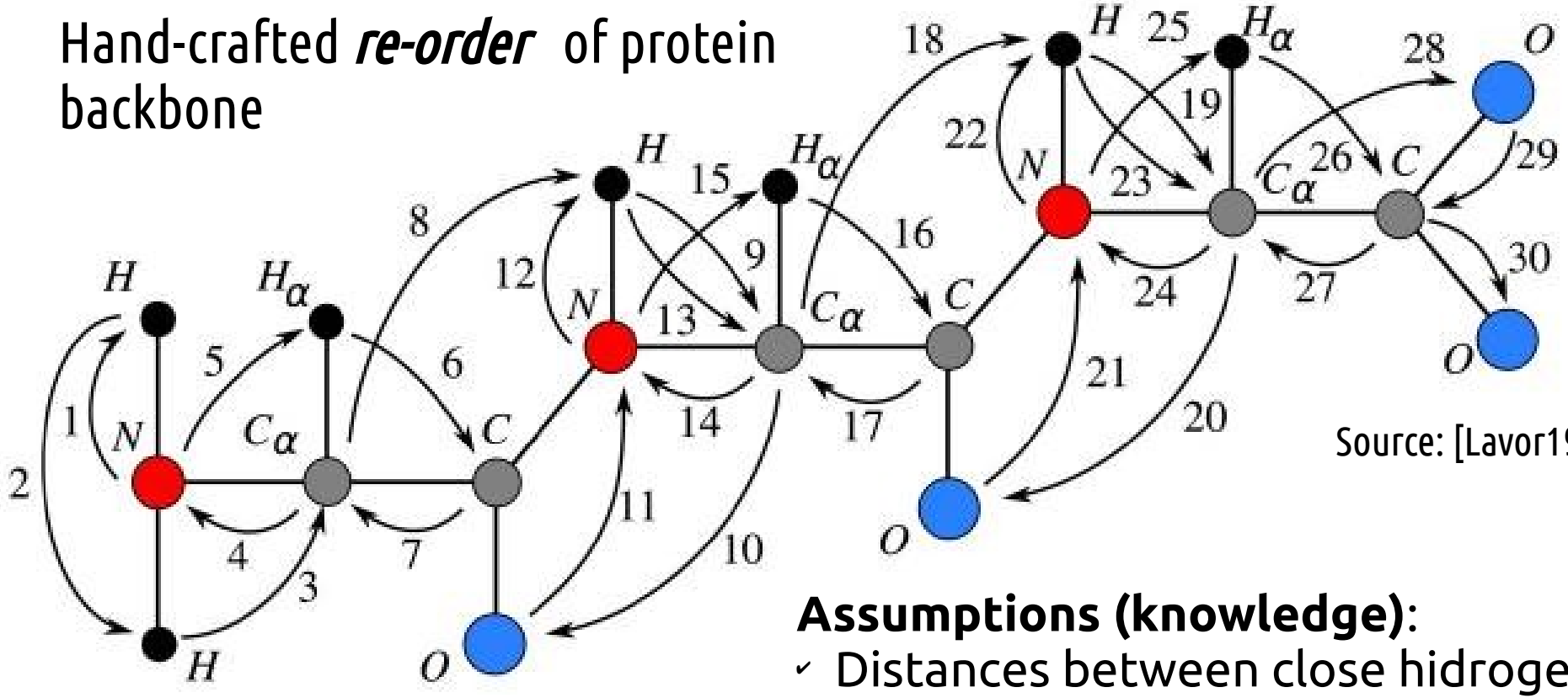
2D

Dependent

A set with more than k points in IR^k has internal orientation (3D chirality)

Does DMDGP$_k$ have any application?

Hand-crafted ***re-order*** of protein backbone



Source: [Lavor19]

Assumptions (knowledge):

- ✓ Distances between close hidrogen atoms are known.

$DMDGP_k$ re-order

Given a undirected graph G=(V, E), a $DMDGP_k$ re-order is a surjetive function r: { 1, …, m } → V, with length m in IN bounded by polynomial in |V|, such that

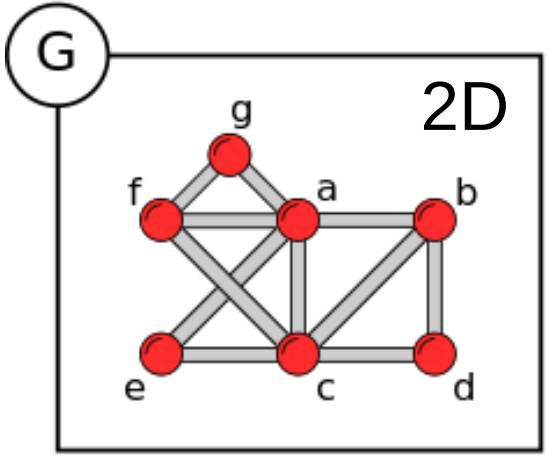
1. { r(1), …, r(k) } defines a k-clique in G;

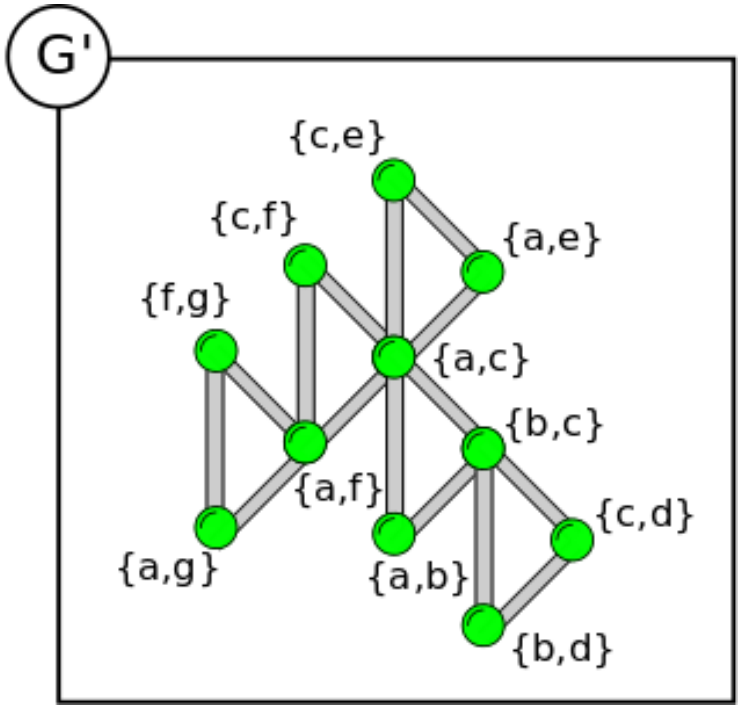
2. For i in { k + 1, …, m }, { r(i – k + j), r(i) } in E for 0 < j < k.

3. For i in { k + 1, …, m }, { r(i – k), r(i) } in E or r(i – k) = r(i)

If there is a re-order, then the discrete search space and symmetry properties are valid

How hard is to check if there is a re-order (k fixed)?



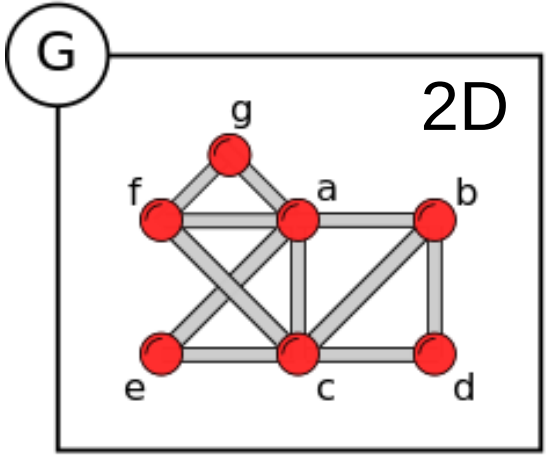


1. Creates G' from G.
2. Does G' have a component H whose the union of its vertices is V?

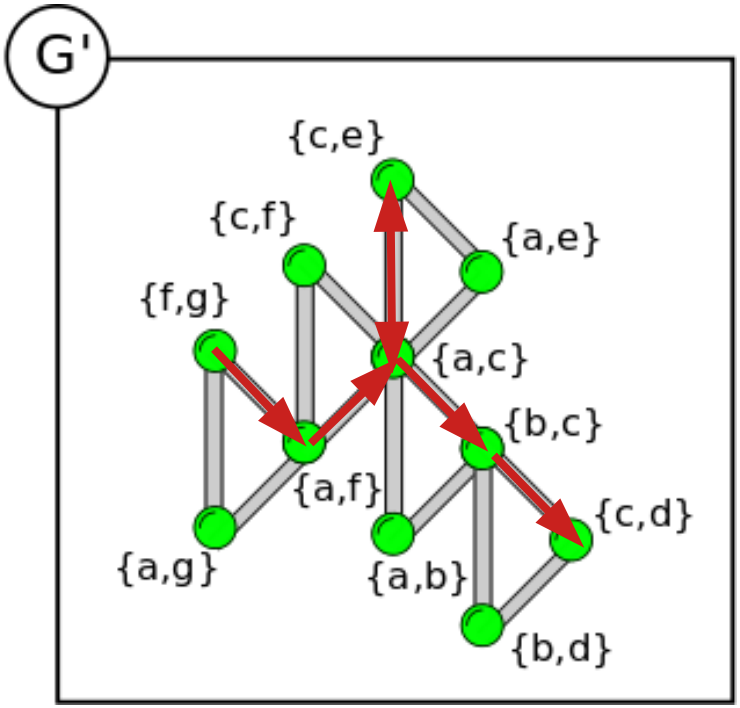
Decision problem

$O(k^3|V|^{k+1})$ steps

How hard is to check if there is a re-order (k fixed)?



(f,g,a,c,e,c,b,d)



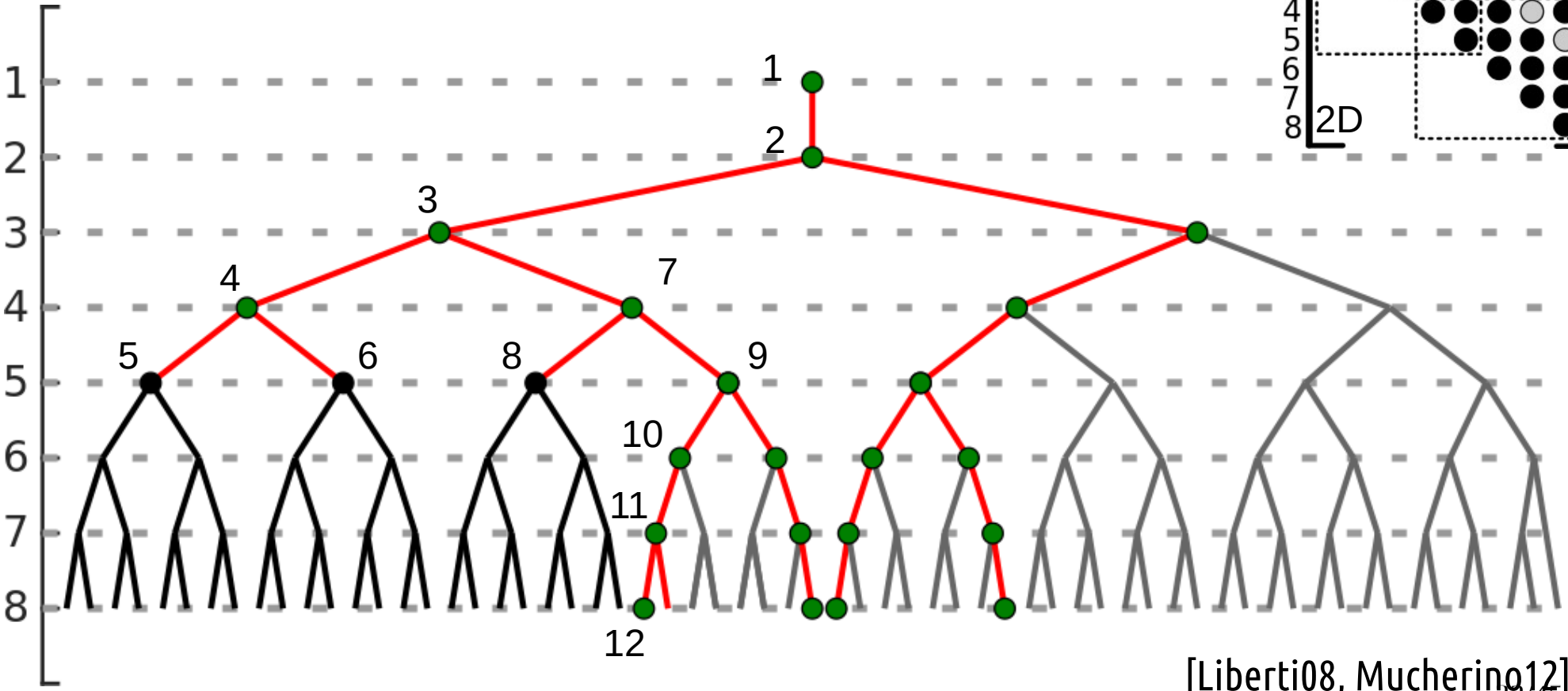
1. Creates G' from G.
2. Does G' have a component H whose the union of its vertices is V?

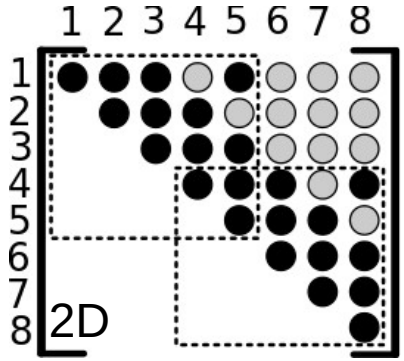
3. Creates a re-order from any walk on H.

Finding a re-order

$O(|V|^{2k})$ steps

Branch-and-Prune (BP) algorithm





[Liberti08, Mucherino12]

Branch-and-Prune (BP) algorithm

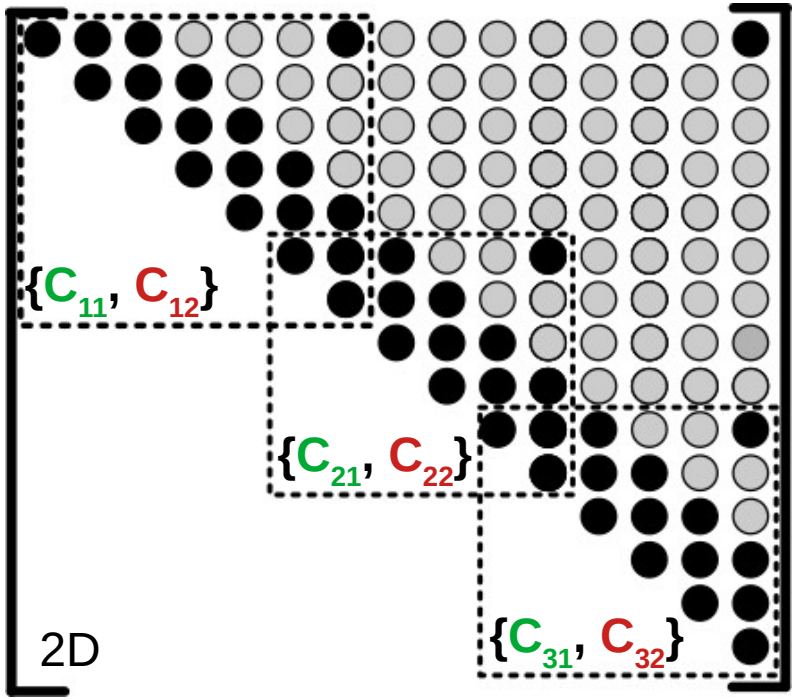
1. Explores all branches (one at time)

2. Prunes the infeasible ones ASAP

3. When a one solution is found, BP generates all the other by flipping the IC's.

Improving BP (SBP)

BP remembers its mistakes (prunning), but not its right choices.



- **Amount of calculations (worst cenario)**

  Let say that solutions are

  $$C_{11} + C_{22} + C_{32} \text{ and } C_{12} + C_{21} + C_{31}$$

  BP will generate the following configurations

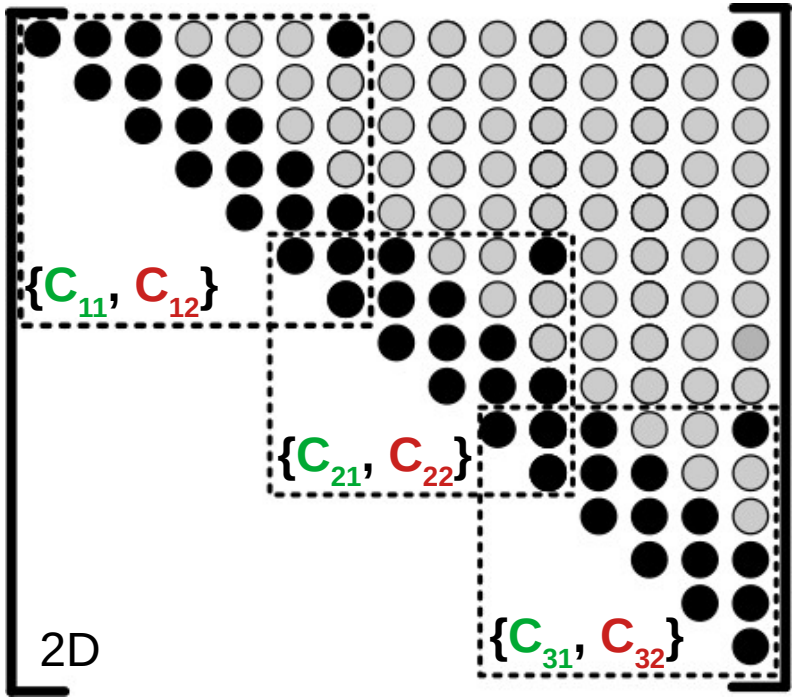
- $C_{11} + C_{21} + C_{31}$ $\quad 2^{|C_1|-k} + 2^{|C_2|-k} + 2^{|C_3|-k}$
- $C_{11} + C_{21} + C_{32}$ $\quad 2^{|C_3|-k}$
- $C_{11} + C_{22} + C_{31}$ $\quad 2^{|C_2|-k}$
- $C_{11} + C_{22} + C_{32}$ $\quad 2^{|C_3|-k}$

$$2^{|C_2|-k} + 2^{|C_3|-k+1}$$

This additional cost can be reduced

Improving BP (SBP)

BP remembers its mistakes (prunning), but not its right choices.



- **Deal with all components as rigid bodies (only reflections are needed)**

  Again, let say that solutions are

  $C_{11} + C_{22} + C_{32}$ and $C_{12} + C_{21} + C_{31}$

  SBP will generate the following configurations

- $C_{11} + C_{21} + C_{31}$ $\quad 2^{|C_1|-k} + 2^{|C_2|-k} + 2^{|C_3|-k}$
- $C_{11} + C_{21} + C_{32}$ $\quad |C_3|$
- $C_{11} + C_{22} + C_{31}$ $\quad |C_2|$
- $C_{11} + C_{22} + C_{32}$ $\quad |C_3|$

$$\boxed{|C_2| + 2|C_3|}$$

$$\boxed{2^{|C_2|-k} + 2^{|C_3|-k+1}}$$

Improving BP (SBP)

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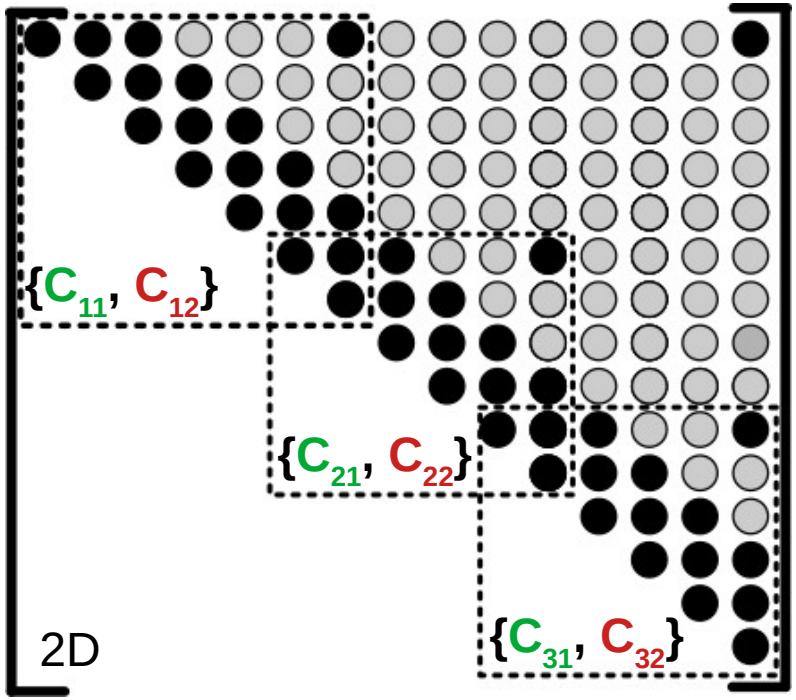

- **Deal with all components as rigid bodies (only reflections are needed)**

  Again, let say that solutions are

  $C_{11} + C_{22} + C_{32}$ and $C_{12} + C_{21} + C_{31}$

  SBP will generate the following configurations

- $C_{11} + C_{21} + C_{31}$ $\quad 2^{|C_1|-k} + 2^{|C_2|-k} + 2^{|C_3|-k}$
- $C_{11} + C_{21} + C_{32}$ $\quad |C_3|$
- $C_{11} + C_{22} + C_{31}$ $\quad |C_2|$
- $C_{11} + C_{22} + C_{32}$ $\quad |C_3|$

$$\boxed{|C_2| + 2|C_3|}$$

This additional cost can be reduced

Improving BP (SBP)

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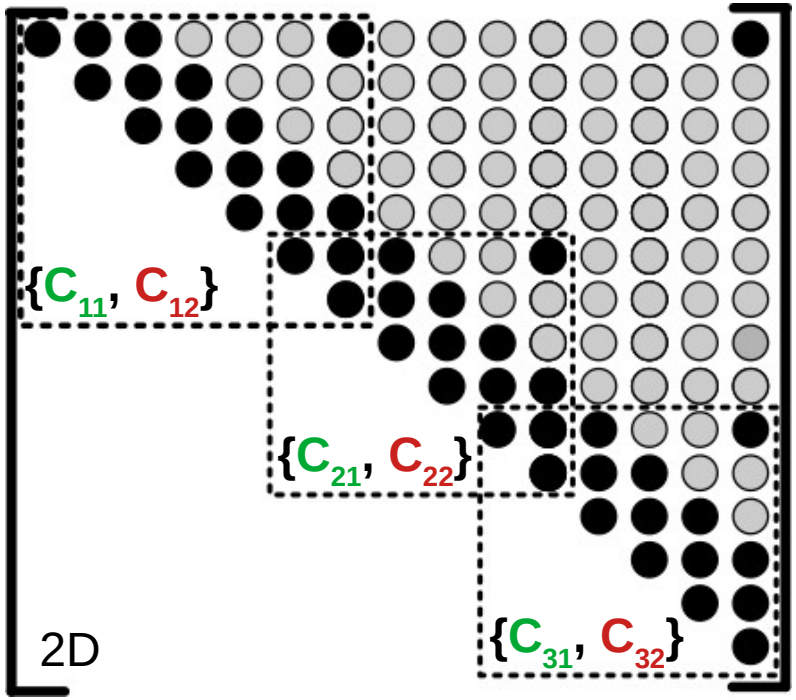

- Deal with all components as a rigid bodies (only reflections are needed)
- **Only the possible locations of the last node need to be tested**

Improving BP (SBP)

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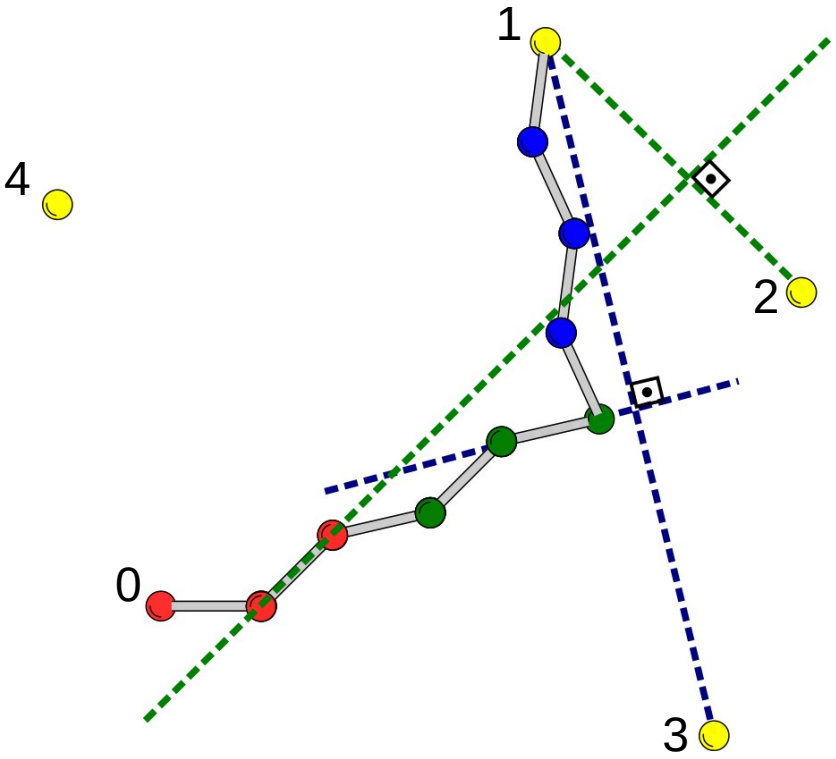

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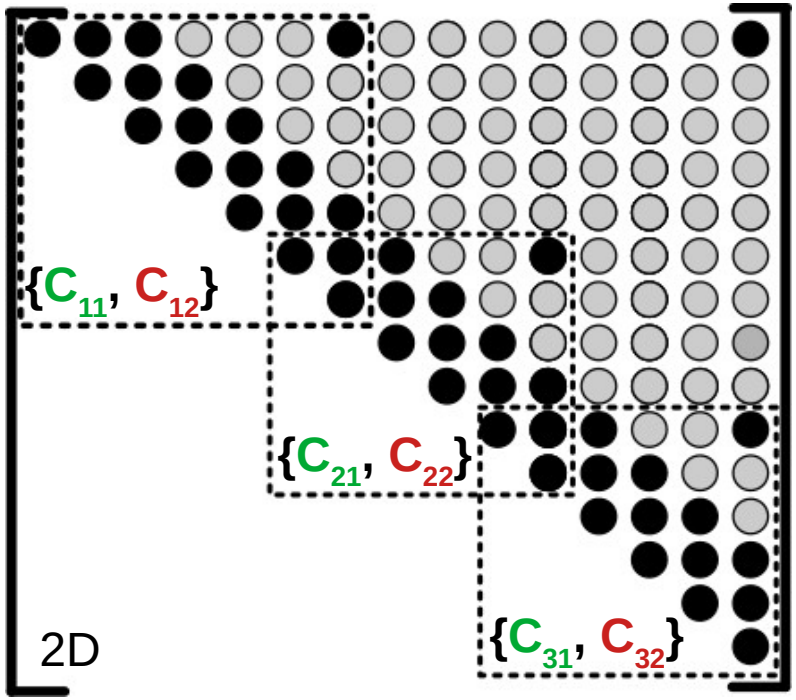


- Deal with all components as a rigid bodies (only reflections are needed)
- **Only the possible locations of the last node need to be tested**

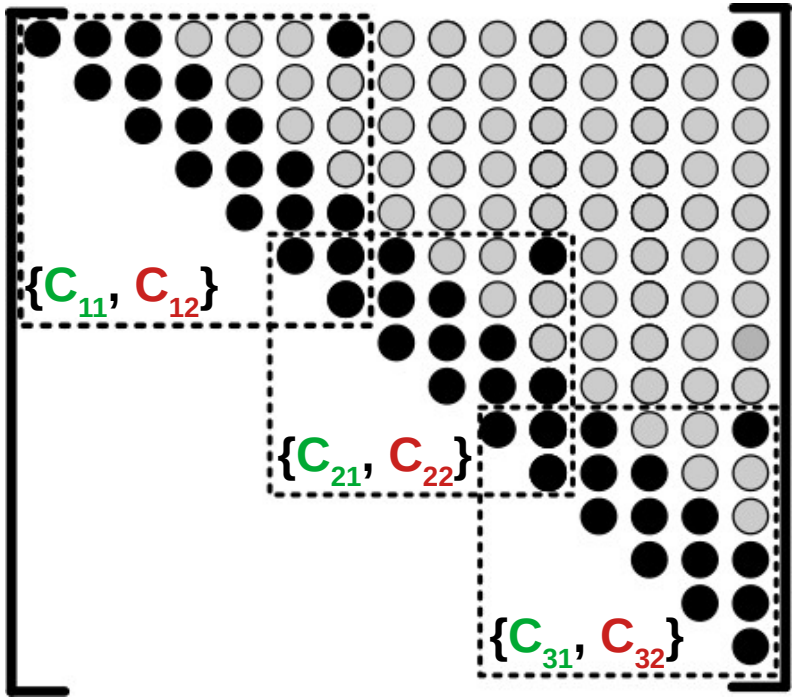
- $C_{11} + C_{21} + C_{31}$ $\quad 2^{|C_1|-k} + 2^{|C_2|-k} + 2^{|C_3|-k}$
- $C_{11} + C_{21} + C_{32}$ $\quad 1$
- $C_{11} + C_{22} + C_{31}$ $\quad 1$
- $C_{11} + C_{22} + C_{32}$ $\quad 1$

$\boxed{1}$

$\boxed{|C_2| + 2|C_3|}$

Improving BP (SBP)

BP remembers its mistakes (prunning), but not its right choices.



- Deal with all components as rigid bodies (only reflections are needed)
- Only the possible locations of the last node need to be tested
- **Merge components**

$$\boxed{2^{|C_1|-k}+2^{|C_2|-k+1}+2^{|C_3|-k+2}}$$

$$\boxed{2^{|C_1|-k}+2^{|C_2|-k}+2^{|C_3|-k}+|C_2|+|C_3|}$$

Numerical Experiments

Implementations of SBP, BP, MDJEEP (V 2.0) [Mucherino19]

Language: C & C++,

Compiler: GCC 7.4.0 (Ubuntu 7.4.0-1ubuntu1~18.04.1)

Intel(R) Core(TM) i3-7100U CPU @ 2.40GHz (16 GB RAM)

Only one solution

Default config of MDJEEP

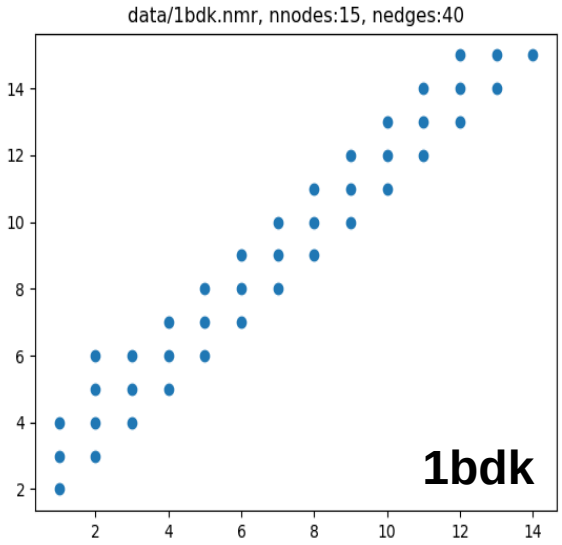
Max time 500 secs

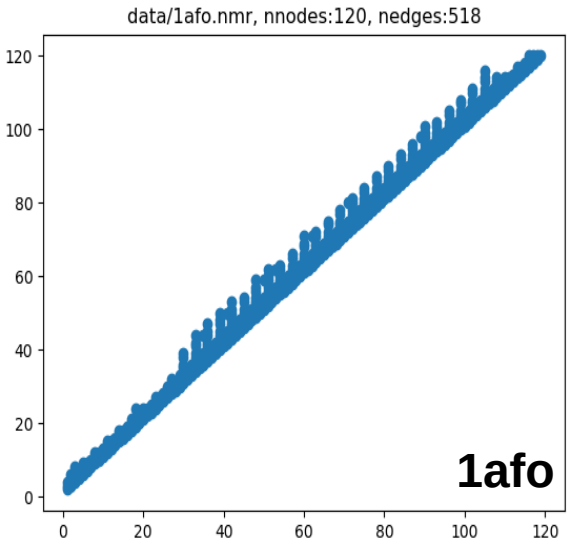
Instances:

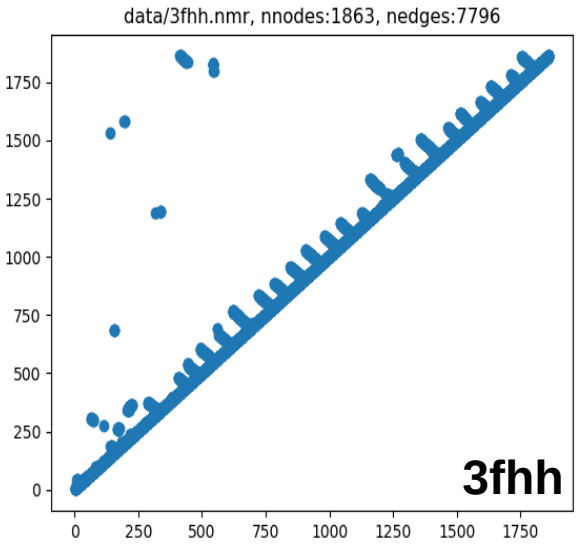
- Artificial
- Only N, C, Ca
- {i, j} in E iff

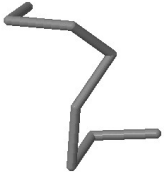
  ((j - i) < 4) or
  d(i, j) < 5 AA and atom(i) = atom(j) = C

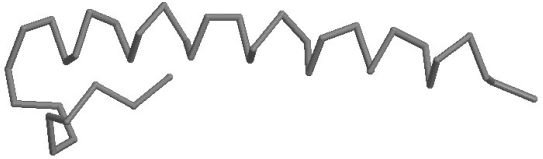
Numerical Experiments











Numerical Experiments

			TIME (secs)			LDE		
fname	natoms	nedges	MDJEEP	BP	SBP	MDJEEP	BP	SBP
1bfw	12	30	1.60E-05	2.60E-06	6.92E-07	2.23E-16	5.58E-16	6.09E-16
1bdk	15	40	2.50E-05	3.51E-06	1.28E-06	2.53E-16	3.75E-16	4.07E-16
1f3r	27	91	5.70E-05	1.58E-05	6.41E-06	2.46E-16	8.87E-16	8.06E-16
1bcv	57	187	9.20E-05	2.03E-05	8.55E-06	3.75E-06	2.18E-13	2.96E-13
1afo	120	518	1.99E-04	3.47E-05	2.42E-05	1.82E-06	2.23E-12	1.97E-12
1b4r	240	1016	4.97E+00	6.80E-01	9.52E-04	5.79E-14	2.19E-12	3.39E-12
1a3s	474	2043	1.84E+01	3.61E+00	1.74E-03	5.25E-07	9.28E-11	2.13E-11
1all	480	2199	1.06E-01	1.28E-02	1.72E-04	1.98E-07	9.02E-13	2.45E-12
1bob	918	3858	1.48E+02	7.36E+00	1.46E-02	1.48E-13	1.86E-11	2.38E-11
3fhh	1863	7796	--	--	2.66E-01	-1.00E+00	-1.00E+00	6.50E-11

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[Lavor19] Lavor, C., Souza, M., Carvalho, L. M., & Liberti, L. (2019). On the polynomiality of finding ^{K}DMDGP re-orders, to appear.